VDS2526 Project Design Report

Part 1. Metadata

- Version: 2.0
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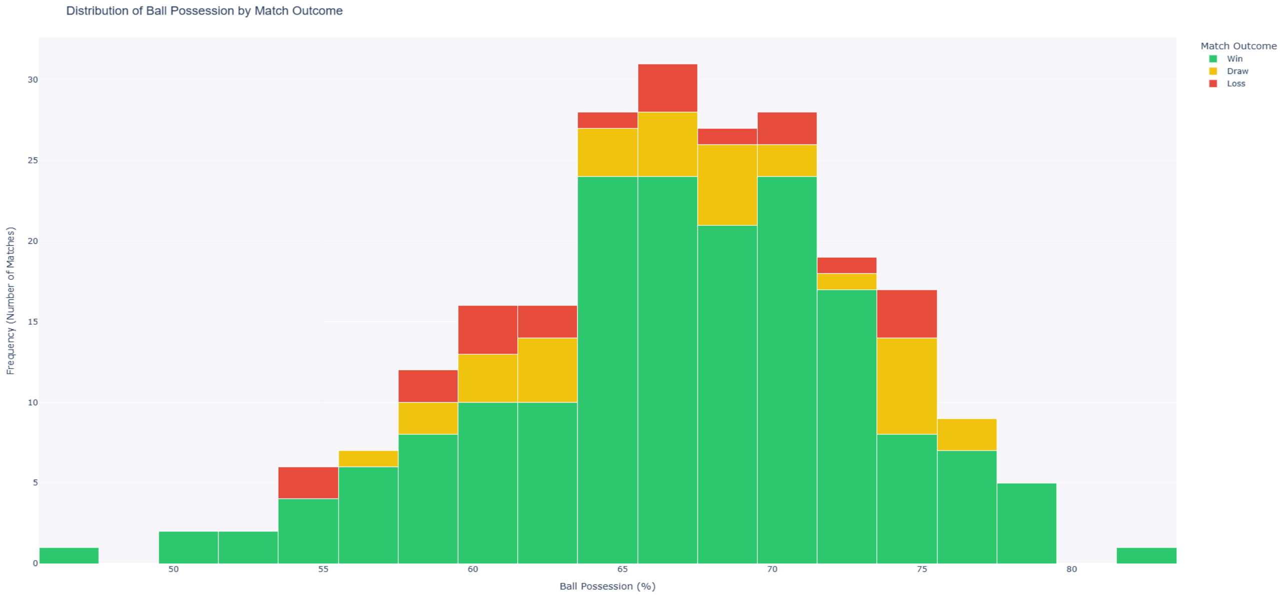
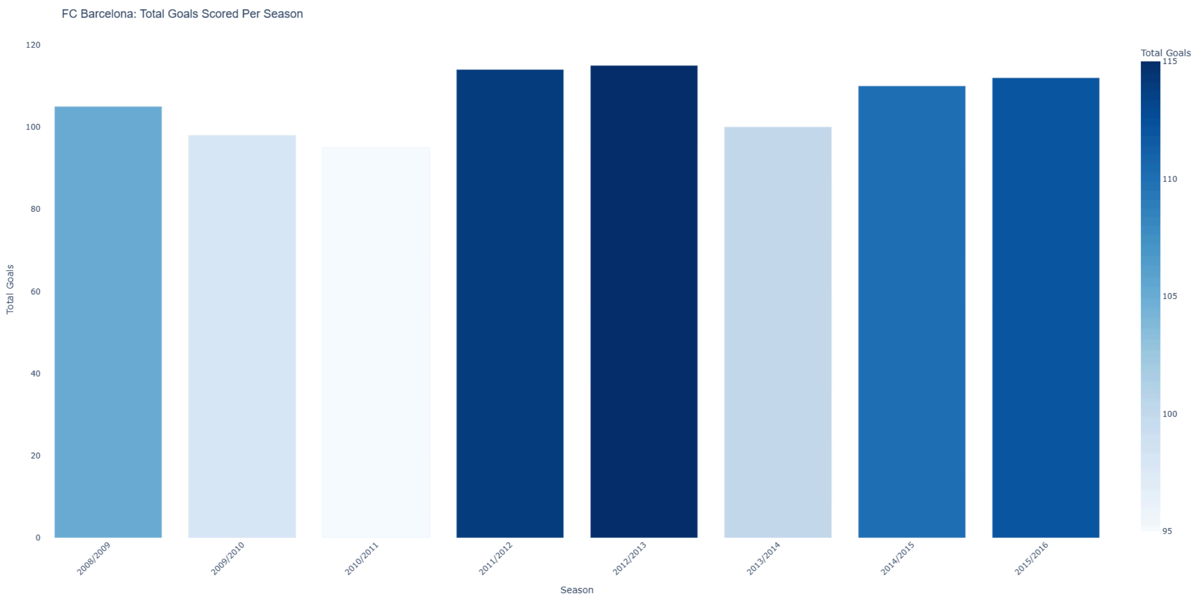
  2502333)
- Group number: group_3
- Dataset: Football

Part 2. Project description

- **<u>Description of Data</u>**
  - Using data, this project aims to assist FC Barcelona decide on its strategy and transfer policy for the next season. To do this, we will look at a large dataset of football games from 2008 to 2016, including their league games, team performances, and detailed player stats. The statistical insights will be used to help the manager's vision by showing them to the club board in a way that makes sense.
- **<u>Exploratory analysis of the data</u>**:
  - **Dimensions and Connections**: We filtered the central Match table exclusively for FC Barcelona. This primary hub connects to the Team dimension via API IDs for match outcomes, and links to the Match_Possession and Match_Goals tables via unique match IDs to integrate tactical and offensive data.
  - **Summary Statistics:** From 2008 to 2016, FC Barcelona played **306** matches. Our analysis reveals an average ball possession of **66.72%** and an impressive average of **3.26** goals scored per home game.
  - **Simple Visualisation:** We used a **bar chart** to display total goals scored per season, and a **histogram** to show the distribution of ball possession percentages across different match outcomes.



- **Total Goals Per Season (Bar Chart):** FC Barcelona were always consistent with their goals between 2008 and 2016, never falling short of the 95-goal mark in any of the seasons. The years 2011/2012 and 2012/2013 were their peak seasons, scoring 110 goals in total in each of these campaigns.



- **Ball Possession vs. Match Outcome (Histogram):** Possession is usually between 60% and 75%, which is a very high percentage of wins, but the most important thing to learn is that losses can occur at nearly any possession figure over 70%. This is already a clear example that possession is not equal to success, and it's time for our deeper tactical analysis.

- **Guiding Questions:**
    - Tactical Efficiency: How do specific match events (e.g., ball possession percentages and types of goals scored) correlate with our win rate in away and home matches?
    - Transfer Strategy: Based on our team's tactical profile, which top 10 under-25 players in other European leagues show the most consistent growth in important FIFA skills like short passing and vision? These players should be targeted for future bookings.
    - Performance Evolution: What specific opponents or match situations have caused our offensive output (goals and shots) to go down over the past few seasons? This will help us find the areas where the team needs a boost.
    - Comparative benchmarking: How does FC Barcelona’s playing style (possession, shots, and goal efficiency) compare to other top European teams, and what differences explain variations in success?

Part 3. Design

3.1 Operationalisation: from questions to tasks

Task 1

- **Title:** Tactical Reality Check (Possession vs. Win Rate)
- **Question**: Does dominating ball possession actually guarantee us wins in away and home games?
- **Action**: Compare and Correlate
- **Target**: Distribution and median of ball possession percentages
- **Objects**: FC Barcelona matches (Home & Away to establish a baseline for away performance)
- **Measure**: Final ball possession % (max elapsed time)
- **Groupings**: Match Outcome (Win, Draw, Loss) & Venue (Home/Away)

Task 2

- **Title:** Scouting for Hidden Gems (U-25 Player Progression)
- **Question**: Who are the young talents in other European leagues (under 25) whose technical skills have been steadily improving?
- **Action**: Identify and Track
- **Target**: Positive upward trends in technical attributes over the last 3 seasons
- **Objects**: Football players (specifically under 25 years old)
- **Measure**: Player age, Short passing score, Vision score
- **Groupings**: Player name, Date (Season)
- **Definitions**:
    - *Steadily improving:* Defined as a consecutive year-to-year increase in skill scores over 3 seasons.

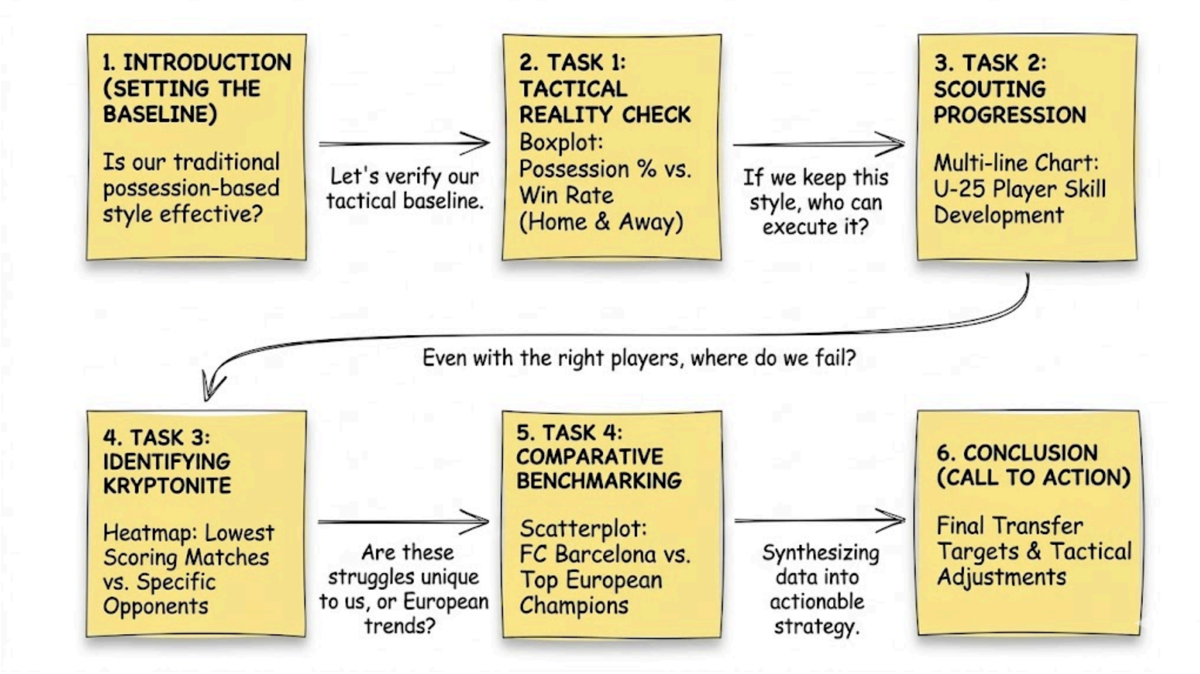
Task 3

- **Title:** Identifying Our Kryptonite (Opponent Vulnerability)
- **Question**: Which specific opponents have figured out how to shut down our attack in recent years?
- **Action**: Discover
- **Target**: Lowest scoring averages (hot/cold spots) against specific opponents
- **Objects**: FC Barcelona matches
- **Measure**: Total or average goals scored by FC Barcelona
- **Groupings**: Opponent team name, Season
- **Definitions**:
  - *Shut down attack:* Defined as FC Barcelona scoring less than 1 goal in a specific match.

Task 4

- **Title:** Comparative Benchmarking (Style vs. Success)
- **Question:** Does our possession-based style lead to better results compared to other top European teams?
- **Action**: Benchmark and Compare
- **Target**: Correlation between high possession and positive match outcomes
- **Objects**: Matches of FC Barcelona and selected top European teams
- **Measure**: Average possession percentage, Win rate, Goal efficiency
- **Groupings**: Team
- **Definitions** :
  - *Top European teams:* Defined as the league champions of the top 5 UEFA leagues (e.g., Real Madrid, Bayern).

3.2 Storyboarding



3.2.1 Storyboarding Overview & Task Execution Plan

We used a sticky-note board to create a sequential storyboard for our workflow. This is our data-driven story line, from establishing a baseline to making actionable transfer and tactical decisions.

1. Introduction (Setting the Baseline)

- Establish the premise: FC Barcelona is known for a possession-based style. Is this traditional style still effective?

2. Task 1: Tactical Reality Check

- **Goal:** Verify our tactical baseline.
- **Execution:** Merge Match and Possession datasets, filter matches, and create a 'Match Outcome' variable.
- **Visual:** Code a **Boxplot** to display the distribution of Possession % versus Win Rate to see if dominance actually guarantees wins.

3. Task 2: Scouting Progression

- **Goal:** If we keep this style, who can execute it in the future?
- **Execution:** Filter the Player dataset for talents under 25 and extract attributes like Short Passing and Vision over time.
- **Visual:** Code a **Multi-line Chart** to track U-25 player skill development and identify future transfer targets.

4. Task 3: Identifying Kryptonite

- **Goal:** Even with the right players, where do we fail?
- **Execution:** Group matches by specific opponents and seasons to find anomalies in our attacking output.
- **Visual:** Code a **Heatmap** to highlight lowest-scoring matches against specific opponents, revealing structural vulnerabilities.

5. Task 4: Comparative Benchmarking

- **Goal:** Are these struggles unique to us, or part of broader European trends?
- **Execution:** Reconstruct season standings across the top 5 major European leagues and isolate the top 3 teams from each.
- **Visual:** Code a **Bubble Chart**. The X-axis represents Average Possession (%), the Y-axis represents Goals per Match, and the bubble size indicates Points per Match. FC Barcelona is highlighted to see where they sit among Europe's elite.

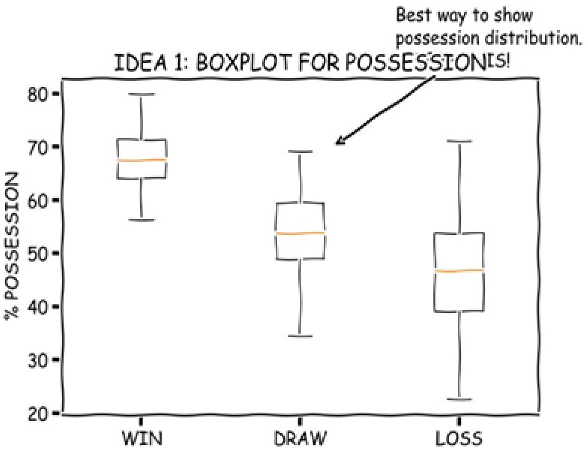
6. Conclusion (Call to Action)

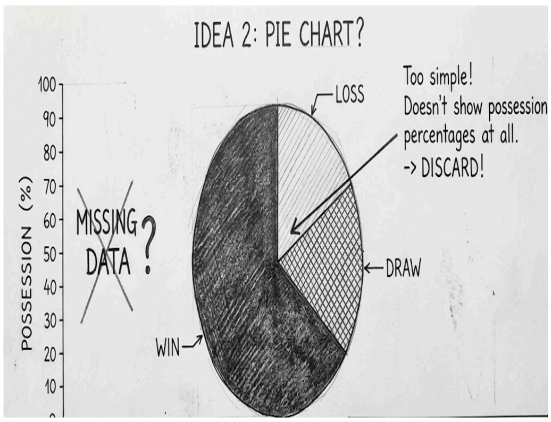
- Combine findings from all tasks into a plan of action and complete tactical changes and transfer objectives.

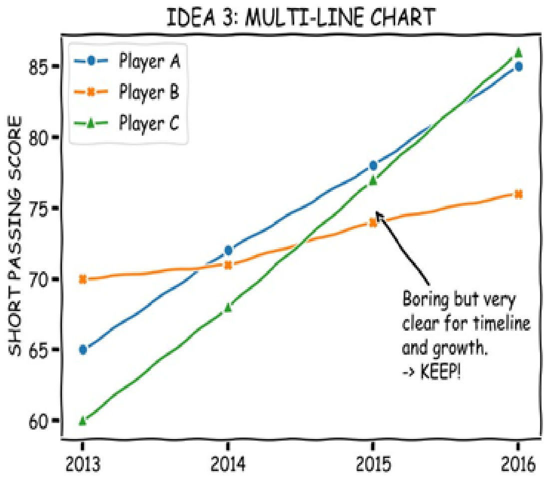
3.3 The design process

3.3.1 Diverge Phase

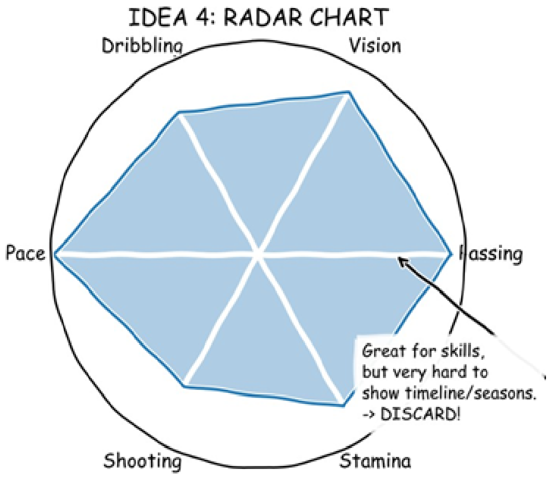
During the diverge phase, each group member independently created between five and ten quick sketches. The focus was on generating diverse concepts rather than refining a single solution, resulting in multiple approaches for each guiding question, including boxplots, scatterplots, heatmaps, and line charts. Each sketch was annotated with axes, variables, and a brief explanation of its purpose.



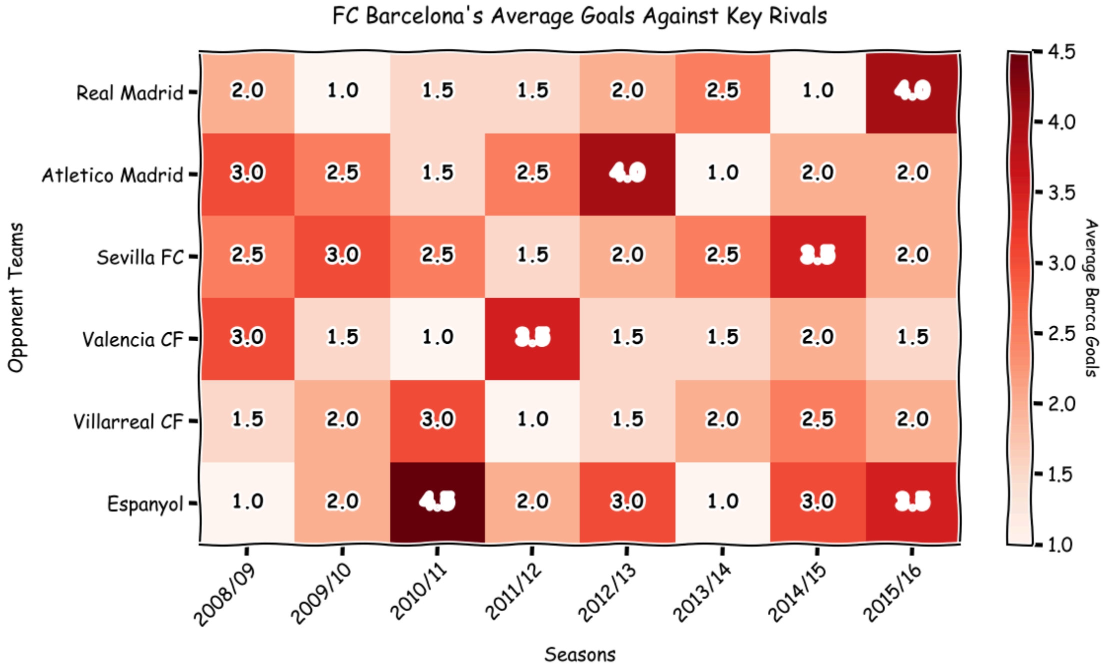




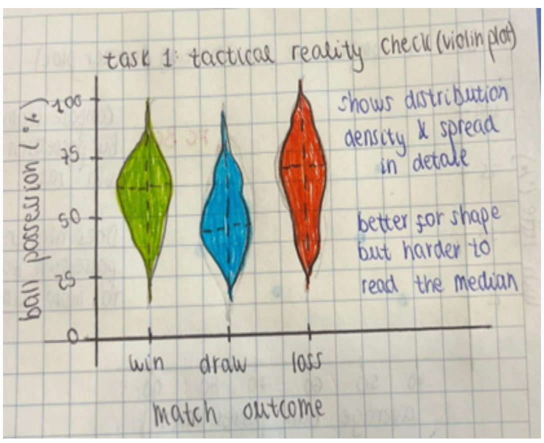
IDEA 3: MULTI-LINE CHART
Player A
Player B
Player C
SHORT PASSING SCORE
60
65
70
75
80
85
2013
2014
2015
2016
Boring but very clear for timeline and growth.
-> KEEP!



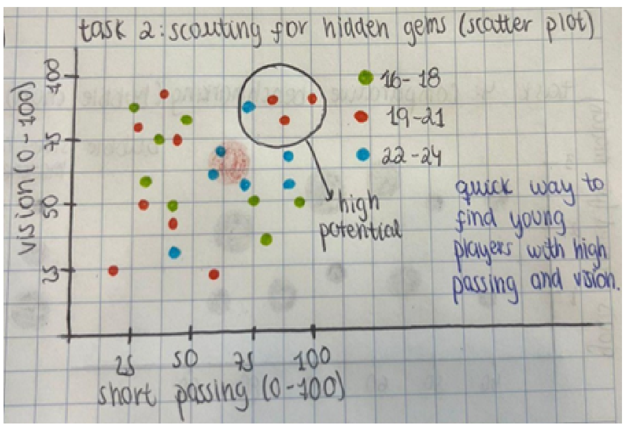
IDEA 4: RADAR CHART
Dribbling
Vision
Pace
Passing
Shooting
Stamina
Great for skills, but very hard to show timeline/seasons.
-> DISCARD!



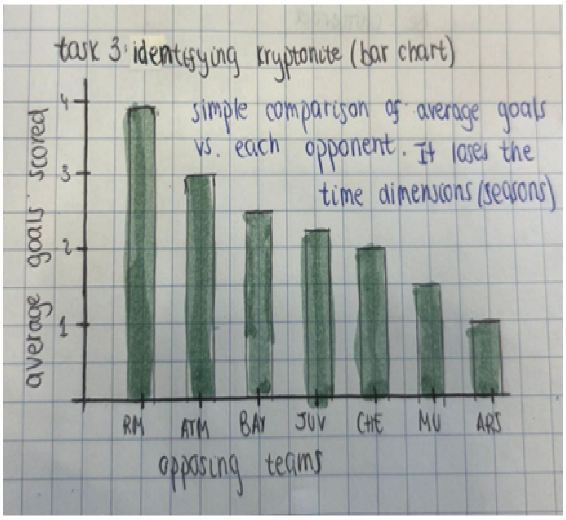
FC Barcelona's Average Goals Against Key Rivals
Opponent Teams
Real Madrid
Atletico Madrid
Sevilla FC
Valencia CF
Villarreal CF
Espanyol
2.0 1.0 1.5 1.5 2.0 2.5 1.0 4.0
3.0 2.5 1.5 2.5 4.0 1.0 2.0 2.0
2.5 3.0 2.5 1.5 2.0 2.5 3.5 2.0
3.0 1.5 1.0 3.5 1.5 1.5 2.0 1.5
1.5 2.0 3.0 1.0 1.5 2.0 2.5 2.0
1.0 2.0 4.5 2.0 3.0 1.0 3.0 3.5
2008/09
2009/10
2010/11
2011/12
2012/13
2013/14
2014/15
2015/16
Seasons
Average Barca Goals
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5

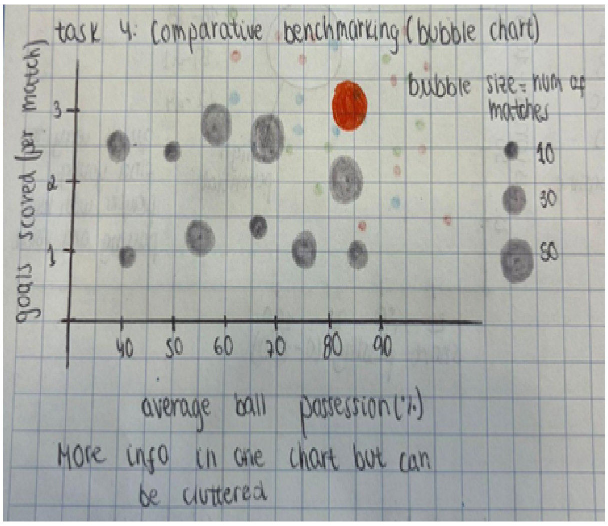


task 1: tactical reality check (violin plot)
ball possession (%)
100
75
50
25
0
win
draw
loss
match outcome
shows distribution density & spread in detail
better for shape but harder to read the median



task 2: scouting for hidden gems (scatter plot)
vision (0-100)
100
75
50
25
16-18
19-21
22-24
high potential
quick way to find young players with high passing and vision.
25
50
75
100
short passing (0-100)





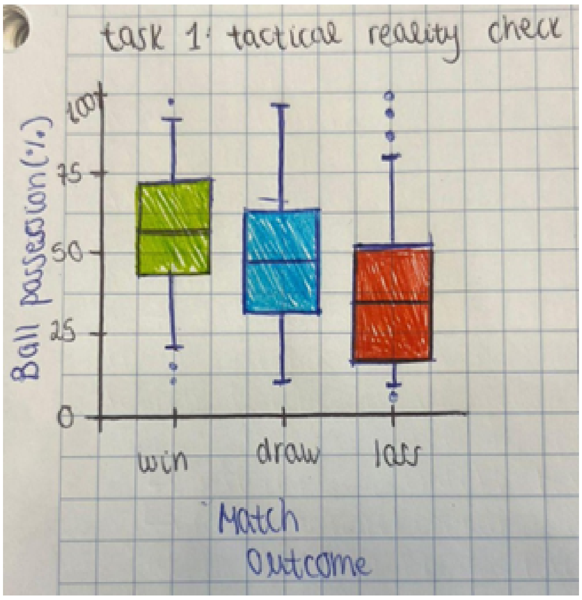
3.3.1 NUF Matrix

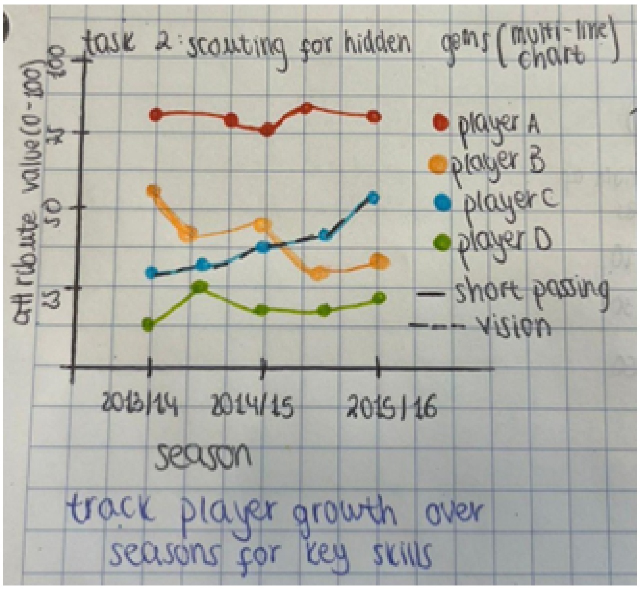
Task / Question	Proposed Visualization Idea	New (1-10)	Useful (1-10)	Feasible (1-10)	Total Score	Status
Task 1: Tactical Reality Check	Boxplot (Possession vs. Outcome)	7	9	10	26	**Selected**
	Pie Chart (Possession Breakdown)	2	3	10	15	Discarded (Fails to show distribution)
Task 2: Scouting Progression	Multi-line Chart (Skill Growth over Time)	6	10	9	25	**Selected**
	Radar Chart (Player Attributes)	8	4	7	19	Discarded (Hard to track changes across multiple seasons)
Task 3: Identifying Kryptonite	Heatmap (Opponents vs. Seasons)	8	9	9	26	**Selected**
	Simple Bar Chart (Avg Goals per Opponent)	3	5	10	18	Discarded (Loses the temporal/season dimension)
Task 4: Comparative Benchmarking	Context vs. Focus: Bubble Chart over Scatter Plot	9	10	8	27	**Selected**
	Standard Bubble Chart (Only Elite Teams)	4	6	10	20	Discarded (Loses the broader European context)

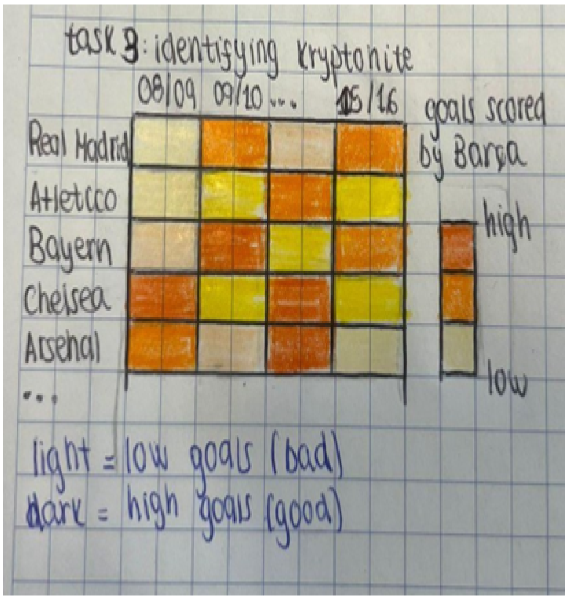
3.3.2 Emerge Phase

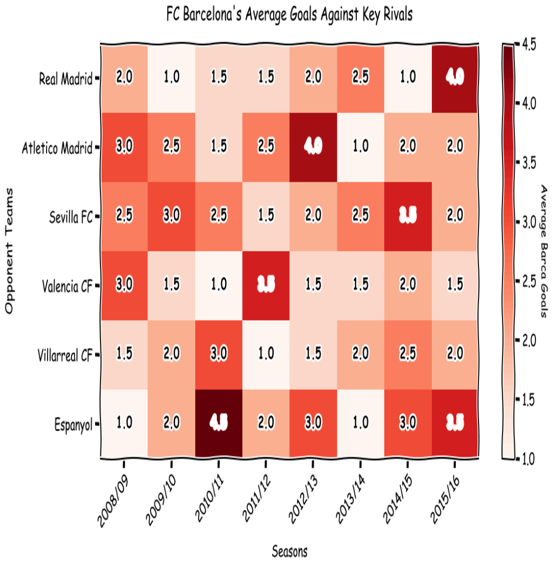
During the emerge phase, we evaluated our 8 design candidates using a New, Useful, Feasible (NUF) matrix to filter out sub-optimal choices.

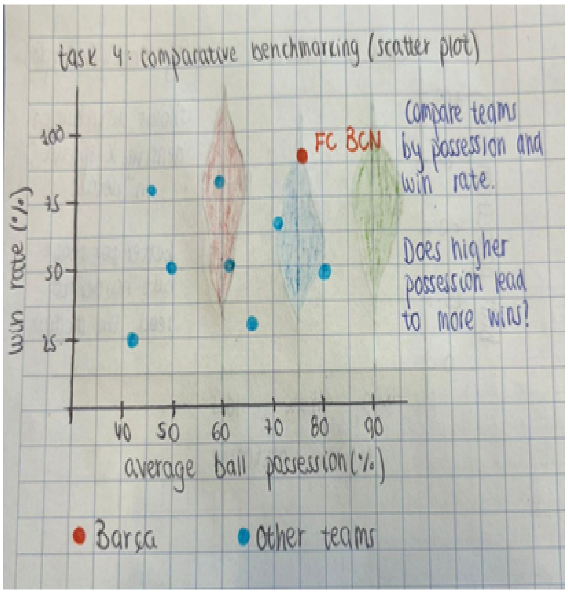
- **Task 1**: We discarded the pie chart because it fails to show distribution data. We chose the boxplot over the violin plot because it provides a much more straightforward way for a general audience to interpret medians and variability.
- **Task 2**: The radar chart was discarded because it is too difficult to track skill changes across multiple seasons. The multi-line chart was selected instead because it is highly effective for visualizing continuous timeline growth.
- **Task 3**: A simple bar chart was rejected because it loses the crucial temporal/season dimension. We selected the heatmap matrix because it preserves both the opponent and time dimensions simultaneously.
- **Task 4**: We preferred a context-focused scatterplot over a bubble chart, as bubble charts introduce too much visual clutter and make clean comparisons difficult.

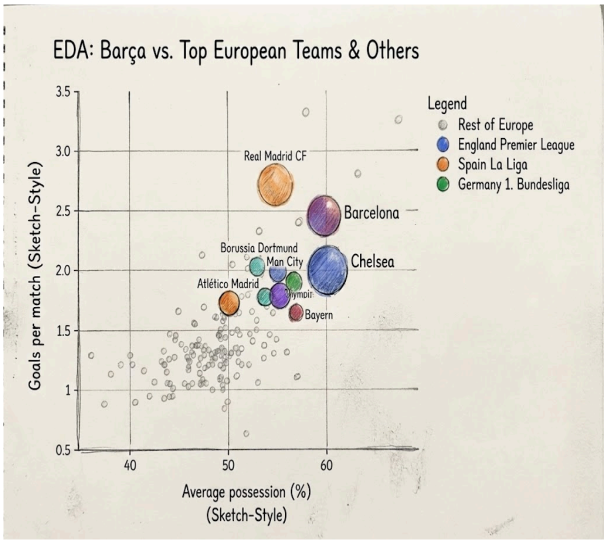












3.3.3 Converge Phase

In the converge phase, we finalized our four selected concepts based on peer feedback and visualization theory. We adjusted our parameters (e.g., standardizing the heatmap color scale and defining top European teams) and refined the sketches to serve as the exact blueprints for our implementation. The final hand-drawn concepts are presented below.

3.4 The final design

We have made some modifications in our designs to make them easy to understand and read after we went from our rough sketches to actual implementation. Here is the breakdown of the visual encodings, design principles and interactions for our final visualizations.

Design 1: Tactical Reality Check (Horizontal Boxplot)

- **Visual Encoding:** During implementation, we realized that a horizontal layout was much easier to read. We mapped the quantitative variable (**Possession %**) to the horizontal X-axis and the categorical variable (**Match Result**) to the vertical Y-axis. We used **boxes**, **whiskers (lines)**, and individual **points** as marks.
- **Design Principles:** By overlaying individual match points over the boxplots, we provide both a clear summary (medians and quartiles) and the actual raw distribution.
- **Interaction:** We included Plotly's native **hover tooltips** to show exact quartile numbers and specific match details without cluttering the screen.

Design 2: Scouting Progression (Multi-line Chart)

- **Visual Encoding:** We mapped **Seasons** to the X-axis and **Attribute Value** to the Y-axis using **lines** and **points**. To solve the problem of showing two different skills per player without creating a mess, we used **color hue** to identify the player and **line style** (solid for short passing, dashed for vision) to identify the skill.
- **Design Principles:** We avoided a massive legend in favor of a clean key on the right. Additionally, we had a strategy to use direct text annotations to identify key data elements (such as highlighting the player who had the steepest growth curve). This helps the viewer get their attention straight to the most actionable insights.
- **Interaction:** The interactive toolbar allows the coaching staff to **zoom in** on specific seasons, **isolate lines**, and read exact scores on hover.

Design 3: Identifying Kryptonite (Heatmap Matrix)

- **Visual Encoding:** We used **areas (grid cells)** as marks, mapping **Seasons** to the X-axis and **Opponent Teams** to the Y-axis. **Color saturation** encodes the average goals scored by Barça.
- **Design Principles:** Incorporating peer feedback, we adjusted the color scale so that darker blue intuitively signifies "more goals" (positive performance). This means that pale/white cells give a natural pop-out effect to have our 'Kryptonite' matches exposed instantly. The design is intentionally filtered along the Y axis to only show the toughest recurring opponents, to reduce visual noise and focus the story.
- **Interaction:** Users can hover over any particular cell to get the precise average number of goals scored against them in a particular season and yet still see the general picture.

Design 4: Comparative Benchmarking (Enhanced Bubble Chart)

- **Visual Encoding:** We upgraded our scatterplot to a bubble chart. **Average Possession** is on the X-axis, and **Goals per match** is on the Y-axis. We added **bubble size** to represent points per match. **Color hue** separates the major European leagues.
- **Design Principles:** So when we were in Europe, and wanted to differentiate FC Barcelona from the others, we decided to give Barça a unique mark, in this case a dark red diamond shape. This has an instant effect and allows the board to instantly see how our team performs and how we compare to the Top 3 teams in each major league for style/efficiency.
- **Interaction:** The chart includes a legend that allows filtering by league and comprehensive hover tooltips to explore individual team stats.

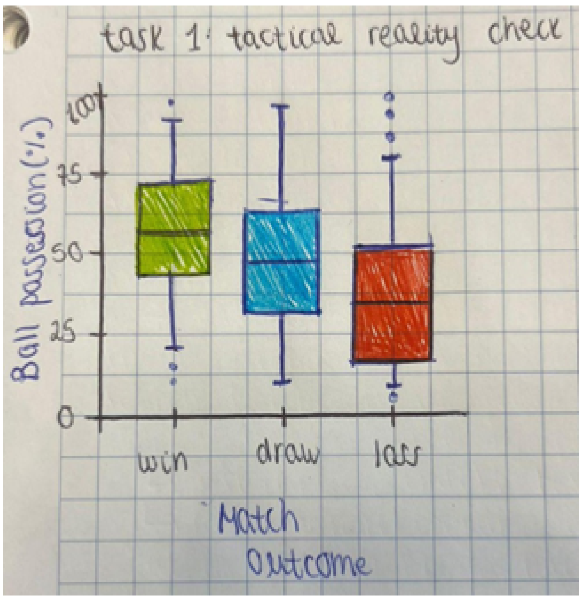
Part 4. Implementation

In this section, you explain the visualisation implementations that you created to answer the questions. For each visualisation, explain the visual encoding and the interactions. Include screenshots (with annotations if relevant).

Please provide the following for each implemented visual:

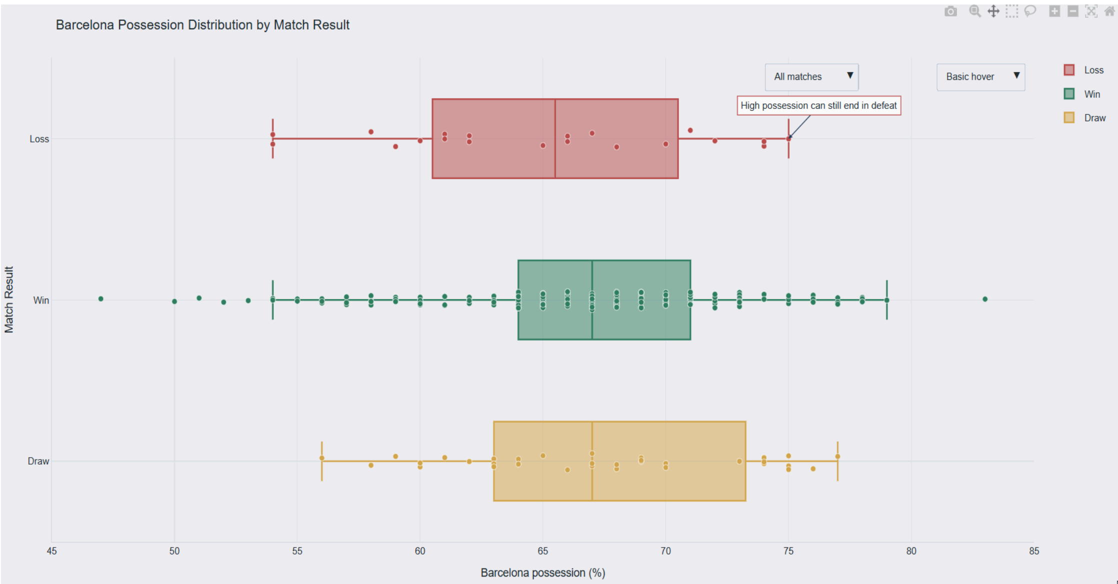
Visualization 1: Possession is not the same as control

- **Intended design**: The intended design for Task 1 was a boxplot comparing Barcelona's possession distribution across match outcomes. The aim was to test whether high possession is consistently associated with winning, or whether Barcelona can still fail to win despite dominating the ball. The boxplot was chosen because it shows median, spread, overlap, and outliers more clearly than a bar chart or pie chart.



- **How the intended design was modified following peer feedback**: Peer feedback suggested that the narrative needed to be clearer and that the story should open with a strong challenge to a common football assumption. For that reason, this visual was kept as the opening panel of the story. The final implementation also simplified the first act by keeping only the boxplot and removing extra visual clutter, so the central message is easier to understand. The explanatory note was also strengthened to explicitly state the intended takeaway: high possession can still end in defeat.

- **Actual implemented design**: The final implemented visual is a horizontal boxplot with overlaid individual match points. The quantitative variable, Barcelona possession percentage, is encoded on the horizontal x-axis, while the categorical variable, match result, is encoded on the vertical y-axis with the categories Loss, Win, and Draw. Color hue is used to distinguish the three result groups. The box-and-whisker marks summarize the distribution of possession for each result category. The box represents the interquartile range, the line inside the box represents the median, and the whiskers show the spread beyond the quartiles. On top of this summary, individual match observations are plotted as points, which makes it possible to see both the aggregated distribution and the underlying raw values at the same time. This combination supports the main analytical question well, because it allows the viewer to see that Barcelona often has high possession even in matches that end in draws or losses. The chart is therefore designed to communicate the core message that possession is an important part of Barcelona's playing style, but it is not the same as control or guaranteed success.



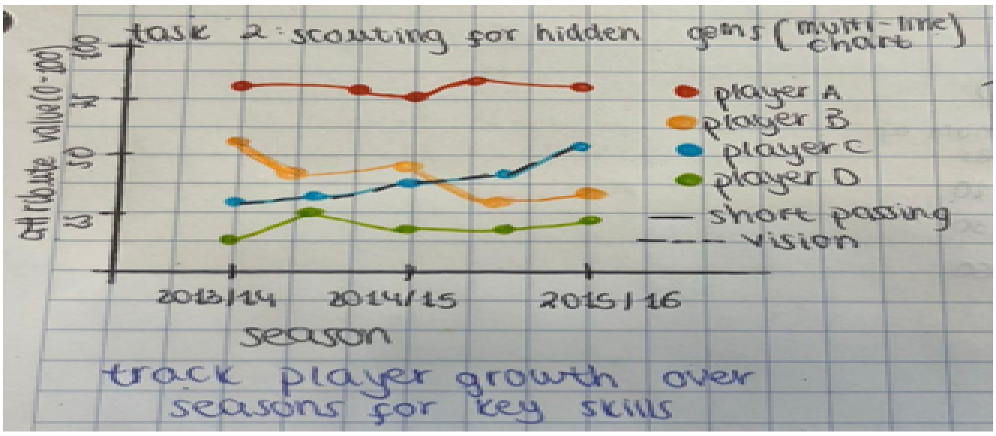
- **Interactions**: The implemented visual includes several interactive features that support exploration without overloading the static view.
    - Hover interaction: when the user hovers over a point or boxplot element, the chart reveals additional contextual information about the match.
    - Hover-detail selector: a dropdown menu allows the user to switch between a more basic hover view and a richer hover view with extra information.

- ○ **Match subset selector**: a second dropdown allows the user to switch between All matches, Home matches, and Away matches, making it possible to compare whether the possession–result relationship changes depending on venue.
- ○ **Legend interaction**: the legend can be used interactively to focus on particular result categories, such as only Wins, only Losses, or only Draws, which helps isolate specific patterns in the data.
- ○ **Navigation controls**: the Plotly interaction tools also support panning, zooming, autoscaling, and resetting the axes, allowing the viewer to inspect parts of the distribution in more detail.
- ○ Scroll-wheel zoom was disabled to make navigation less distracting and to keep interaction more controlled.

These interactions are useful because they let the user move from an overall summary of Barcelona's possession patterns to more detailed inspection of specific match contexts while preserving the main narrative message of the chart.

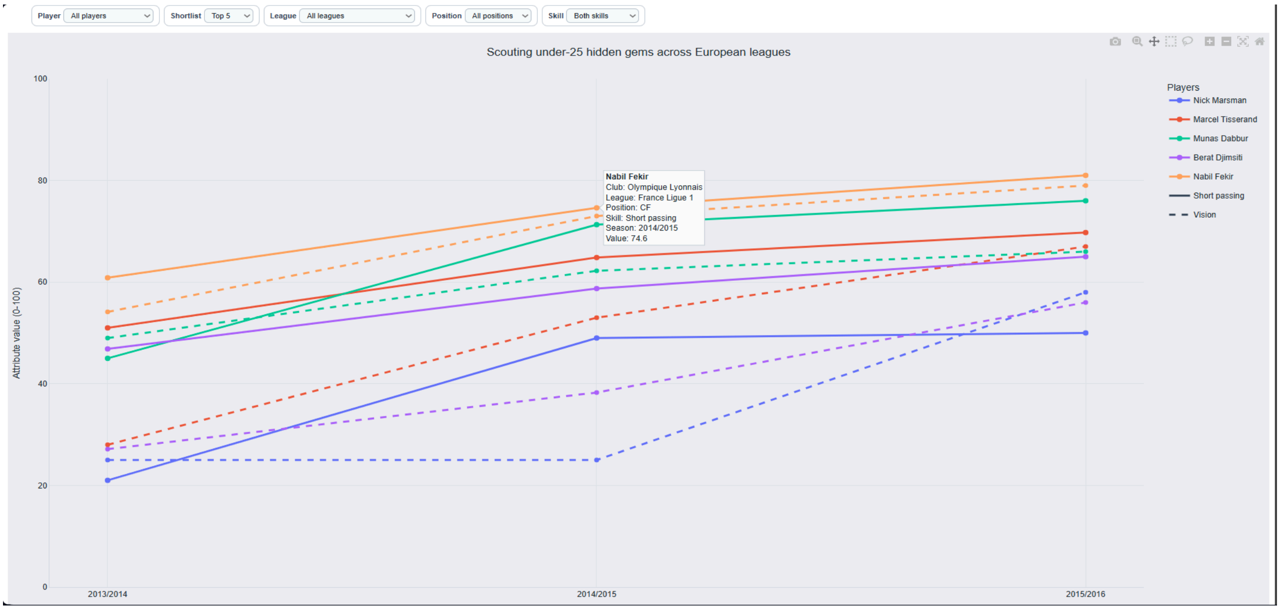
<u>Visualization 2: Scouting for hidden gems</u>

- **Intended design:** The intended design for task 2 was a multi-line chart tracking the development of young players over time. The goal was to identify technically promising players by following the evolution of two attributes, short_passing and vision, across seasons. The line chart was chosen because it is one of the clearest ways to show change over time and compare multiple players' development trajectories.



- **How the intended design was modified following peer feedback**: Peer feedback pointed out that the notion of "consistent development" was too vague in the original proposal. In response, the implemented version operationalises this rule more explicitly: players are connected from Player to Player_Attributes via player_api_id, age is calculated from birthday, only players under 25 are kept, and only players with a positive upward trend in both short_passing and vision over their last three seasons are included. Feedback also warned that a multi-line chart could become overcrowded, so the final version uses consistent color coding, solid versus dashed line styles, and interactive inspection to keep the chart readable.

- **Actual implemented design**: The final visual is a multi-line chart. The marks are lines and point markers. The x-axis encodes the season, and the y-axis encodes the attribute score on a 0 to 100 scale. Color hue identifies each player. Line style distinguishes the two variables: solid lines represent short_passing and dashed lines represent vision. Small point markers indicate observed seasonal values. This encoding makes it possible to compare both within-player development and between-player differences over time. The chart therefore translates the scouting question into a readable comparison of technical growth trajectories.

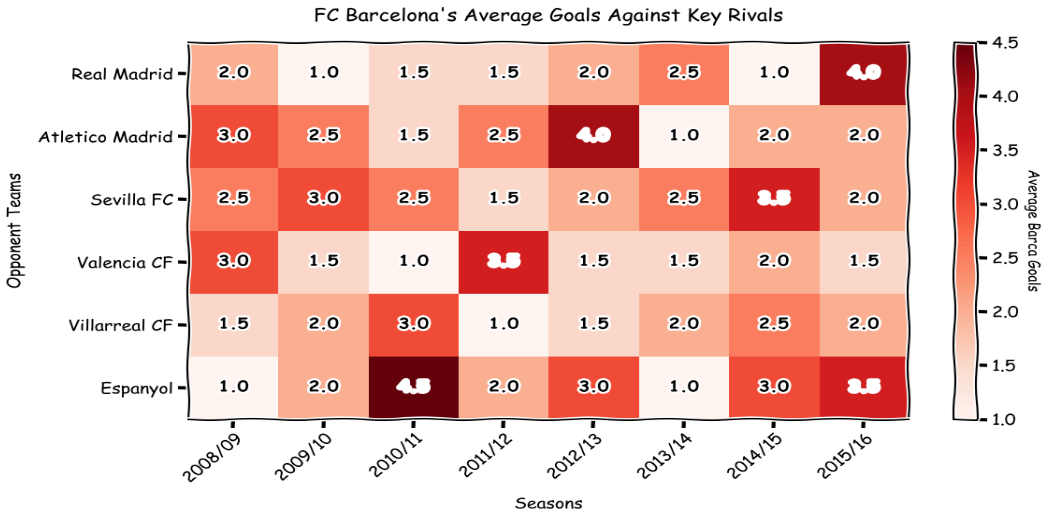


- **Interactions**: The visualization includes several interactions to reduce clutter and support detailed inspection.
  - Player filter: the user can select all players or focus on a specific player.
  - Shortlist filter: the user can choose how many top-ranked players to display, such as Top 5, Top 10, Top 15, or Top 20.
  - League filter: the user can restrict the view to players from a specific European league.
  - Position filter: the user can focus on players from a specific tactical position.
  - Skill filter: the user can show both skills together or isolate only short passing or only vision.
  - Hover tooltips: hovering over a point or line reveals the exact player, club, league, position, season, skill, and attribute value.
  - Legend interaction: the legend can be used to focus on or hide particular player traces.
  - Navigation controls: the chart supports panning, zooming, autoscaling, and resetting the axes.

  These interactions are important because the underlying chart contains many possible lines. Instead of showing everything at once and making the figure difficult to read, the final design follows an overview-to-detail approach: the viewer starts with a manageable shortlist and then filters the data according to the scouting question they want to answer.

Visualization 3: Kryptonite opponents

- **Intended design:** The intended design for Task 3 was a heatmap showing how Barcelona's attacking output changes by opponent and season. The purpose was to identify "kryptonite" opponents: teams that consistently reduce Barcelona's goalscoring across multiple seasons. A heatmap was selected because it allows two categorical dimensions to be compared simultaneously while using color intensity to communicate performance levels.



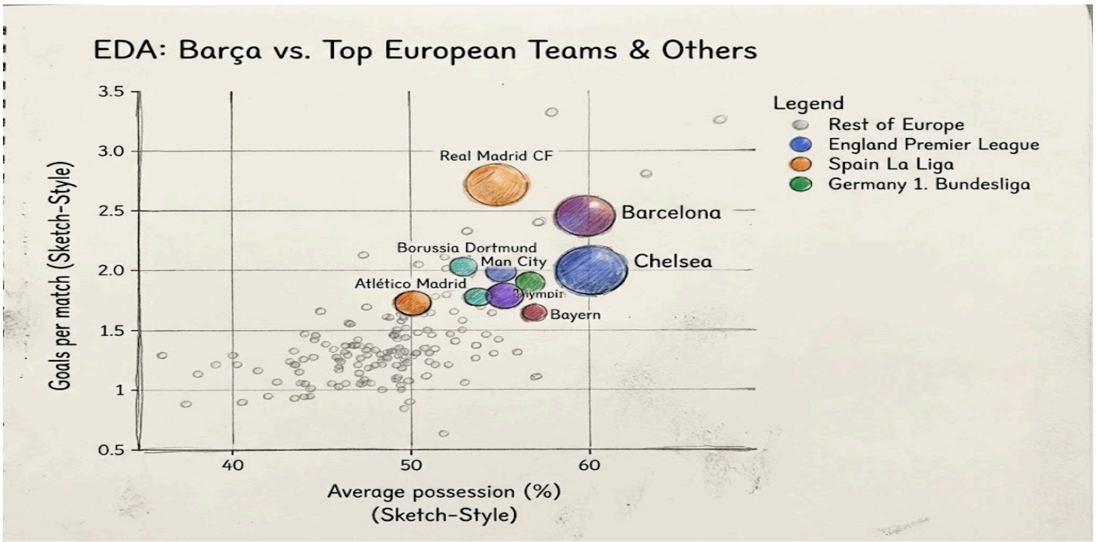
- **How the intended design was modified following peer feedback**: Peer feedback raised two important issues. First, the concept of "shutting down the attack" needed to be made more explicit. Second, the original heatmap risked becoming too crowded and confusing if too many teams were displayed at once. The final implementation addresses this by defining the metric as Barcelona's average goals scored against each opponent in each season, restricting the view to recurring opponents, and then keeping only the 12 toughest opponents by Barcelona's average goal output. The design was also refined so the visual storytelling focuses on sustained weak matchups rather than one-off anomalies.
- **Actual implemented design:** The final visual is a heatmap matrix. The marks are rectangular cells where the x-axis encodes the season and the y-axis encodes the opponent team. Color intensity (using a sequential white-to-dark-blue scale) encodes Barcelona's average goals scored.
  - **Dark blue cells** correspond to higher average goal output (strong attacking performance).
  - **White/very light blue cells** correspond to 0 or very low goals, immediately popping out to indicate our problematic "Kryptonite" matchups.
  - **Empty/grey cells** show that no match occurred between the two teams during that specific season (i.e., they were not in the same league), preventing misleading interpretations. Numeric labels inside the colored cells give the exact average goal value for quick reading.

- **Interactions**: The implemented visualization includes interactions that support closer inspection of the matrix.
    - Hover tooltips: when the user hovers over a cell, the chart reveals the exact season, opponent, and average Barcelona goals for that matchup.
    - Navigation controls: the chart supports panning, zooming, autoscaling, and resetting the axes.
    - Toolbar interaction: standard Plotly toolbar controls allow the user to inspect specific regions of the heatmap more closely.

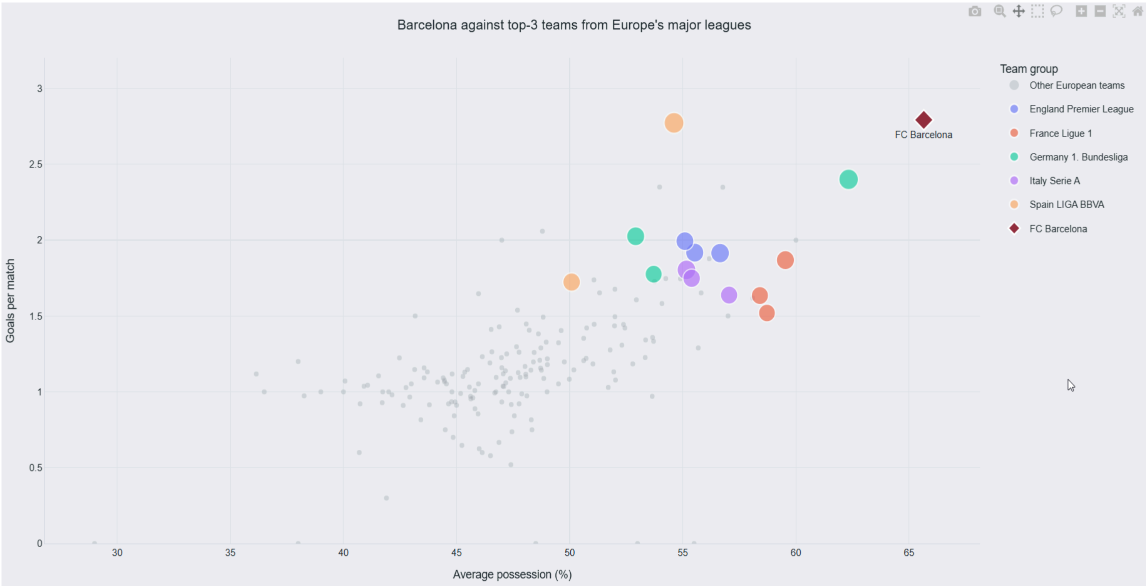
  Because the heatmap already includes direct value labels inside the cells, the main role of interaction is not basic decoding but detailed inspection. It helps the viewer focus on particular opponents or seasons and confirm exact values for the most notable low-scoring matchups.

<u>**Visual 4: Barcelona against Europe's elite**</u>

- **Intended design**: The intended design for Task 4 was a comparative European benchmark showing where Barcelona sits relative to top clubs from Europe's major leagues. The purpose was to place the previous findings in context: if Barcelona sometimes struggles to convert control into output, is that unusual, or is it simply how elite clubs compare? A scatterplot with bubble size was chosen because it allows three quantitative variables to be shown simultaneously in a compact and interpretable way.



- **How the intended design was modified following peer feedback**: Peer feedback pointed out that the idea of "top European teams" was not sufficiently defined in the original concept. In response, the final implementation uses a more explicit data-driven rule: teams are ranked from match results, major leagues are restricted to Spain, England, Italy, Germany, and France, and the benchmark group includes the top three long-run teams from each league. Peer feedback also noted that the chart could become visually crowded, so the final version was cleaned substantially: marker sizes were reduced, label placement was adjusted manually, Barcelona was highlighted only once, and unnecessary label backgrounds were removed to improve readability.
- **Actual implemented design**: The final visual is a scatterplot with variable-sized markers (bubbles). The marks are circular points for benchmark teams and a diamond marker for Barcelona. The x-axis encodes average possession, and the y-axis encodes goals per match. Color hue distinguishes league membership. Direct labels identify the clubs, and Barcelona is emphasized with a distinct shape and color. This encoding supports a comparison of territorial control, attacking output, and competitive strength in one view. It therefore functions as a synthesis visual for the whole story.



- **Interactions**: The visualization includes several interactive features that help the viewer inspect the benchmark more closely.
    - Hover tooltips: when the user hovers over a point, the chart reveals the exact team and its values, such as possession and goals per match.
    - Legend interaction: the legend is interactive, so the user can click league categories to hide or show specific leagues. This is especially useful for isolating one competition at a time or reducing clutter when comparing Barcelona only against a smaller subset of teams.
    - Navigation controls: the chart supports panning, zooming, autoscaling, and resetting the axes.
    - Toolbar interaction: Plotly's toolbar allows the viewer to inspect clusters of nearby teams more precisely.

  These interactions are useful because several benchmark clubs are positioned relatively close to one another. The ability to hide leagues directly from the graph makes the chart easier to explore and helps the viewer answer more specific comparison questions without changing the overall visual design.

In addition, please provide:

1. A link to the code for the visualisations (e.g. the github repository). Make sure you include instructions to reproduce your visualisations. https://github.com/salehsami/Visualisation-in-DS-Project

2. A link to a 3-5 minute youtube video in which you answer the chosen questions by demonstrating the functionality of the visualisations you created. Try to use the data storytelling techniques we saw in class.
   Note: Paste the full URL as links sometimes don't work. For instructions on how to upload a video, see e.g. https://www.youtube.com/watch?v=VtF2AgFSLAw. Important: set visibility to either "public" or "unlisted" (not "private").

   https://youtu.be/V1OkavkefCk
   The video runs around 6 minutes and follows the same narrative order as the web implementation. Suggested structure:
   1. Open with the possession question (Act 1) — show the home/away filter
   2. Transition to the season trends (Act 2) — point out the 2011/2012 peak
   3. Show the heatmap (Act 3) — name Valencia and Hércules specifically
   4. Close with the benchmark (Act 4) — position Barcelona relative to Man City and Bayern